# SMART INDIA HACKATHON 2026

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## CYCLONEX




- **Problem Statement ID –26070**
- **Problem Statement Title-To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-157898**
- **Team Name- DOOMSDAY1**

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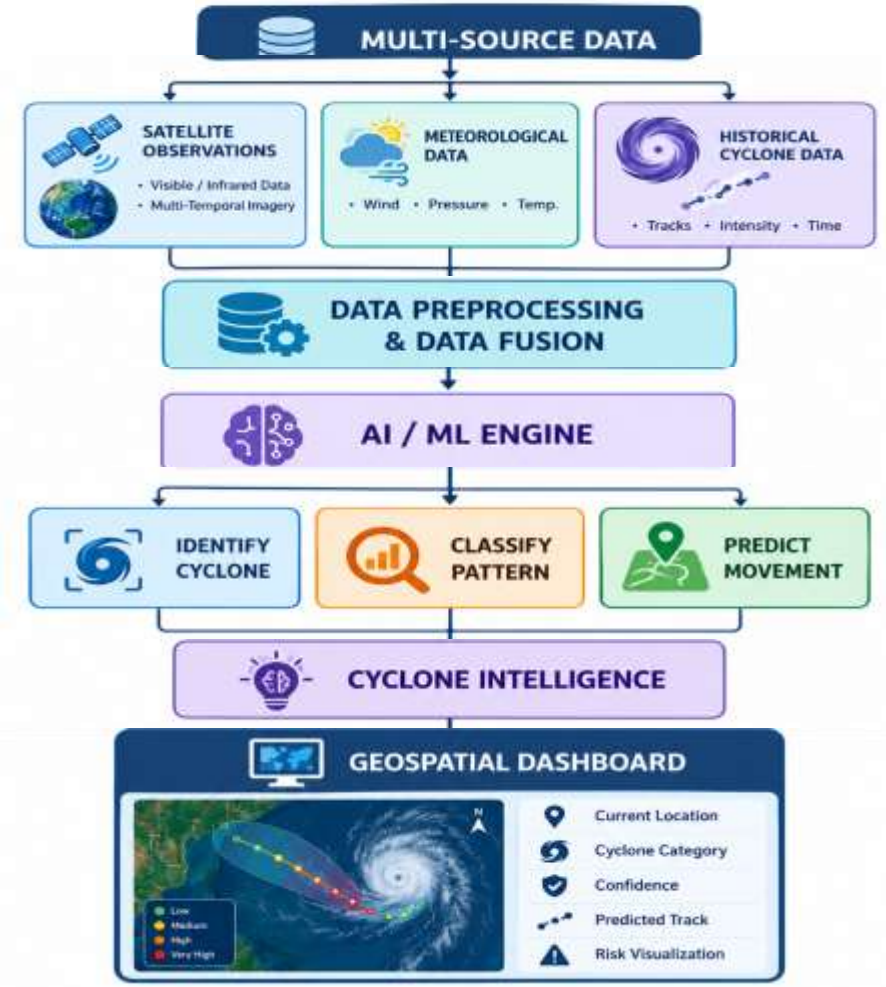
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## ❖ Proposed Solution

- ❖ Detects potential tropical cyclone formations from satellite observations, reducing dependence on purely manual image interpretation.
- ❖ Analyzes visual and meteorological features to categorize the detected cyclone according to learned cyclone characteristics.
- ❖ Uses historical cyclone tracks and temporal observations to estimate the cyclone's future trajectory and evolution.
- ❖ Presents detected cyclones, classification, confidence, location and predicted trajectory through an intuitive geospatial interface.

## ❖Innovation & Uniqueness

- ❖ Instead of relying on a single satellite image, the system is designed to combine **multiple satellite/meteorological data sources** for richer cyclone analysis.
- ❖ One system covers:**Identification → Classification → Prediction → Visualization**
- ❖ rather than solving only one part of cyclone analysis.
- ❖ The system considers both:
- ❖ **WHERE** the cyclone is and
  **HOW it is changing over time.**
- ❖ AI outputs are translated into an understandable **map-based cyclone track**, making complex predictions easier to interpret.




Active Link https://cyclonex-8b1od47ay-harsh-mohans-projects-21df8260.vercel.app/

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# TECHNICAL APPROACH

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### FRONTEND (Next.js + React)

- App Router (SSR)
- TanStack React Query
- Tailwind CSS
- Lucide Icons

### BACKEND (Express.js)

- RESTful API (/v1)
- Data Processing
- Real-time Integration
- Node.js Server

### DATABASE (Prisma ORM)

- SQLite (Dev)
- PostgreSQL (Prod)
- Type-Safe Queries
- Data Seeding

### API ENDPOINTS & INTEGRATIONS

- /cyclones • Trajectory tracks & dossiers
- /telemetry • Global observations & status
- /reports • Tactical intelligence data

### ARCHITECTURE PATTERN

- Monorepo (npm workspaces)
- Client-Server Separation
- Decoupled Architecture
- Real-time Data Sync

### DATA FLOW: Ingestion → API → Caching → Rendering

Meteorological Data (ISRO/IMD) → Express REST API → TanStack Query Polling & Caching → Real-time Dark-Mode Dashboard Rendering



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# FEASIBILITY AND VIABILITY

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## ❖Feasibility:

### Technical Feasibility:

- Proven DL models: **CNN, ConvLSTM, ViT** for cyclone detection, intensity & tracking
- Multi-source satellite & weather data: **INSAT-3D/3DR, Meteosat, GPM, IMERG, ERA5**
- Cloud-based training using **AWS/Colab**
- Web dashboard with **REST APIs**

### Operational Feasibility:

- Supports **IMD & NDMA** workflows
- Automates cyclone analysis and alert generation

### Economic Feasibility

- Open-source ML stack
- Cloud-based deployment reduces infrastructure requirements

## ❖Viability:

### Scalability:

- Can scale from **regional cyclone monitoring to wider coastal coverage**
- Cloud infrastructure supports increasing data volume

### Deployment:

- Web-based dashboard enables centralized monitoring
- REST APIs allow integration with existing systems

### Impact:

- Faster cyclone analysis and early warning support
- Helps reduce risks to **life, infrastructure & property**

## ❖Challenges:

- **Data Imbalance:** Severe cyclones are rare in datasets
- **Data Heterogeneity:** Different resolutions, formats & timestamps
- **Data Gaps:** Cloud cover, sensor outages & missing frames
- **Rapid Intensification:** Sudden intensity changes are difficult to predict
- **Latency:** Large imagery datasets can slow real-time alerts
- **False Alarms:** Incorrect warnings may reduce user trust

## ❖Mitigation Strategies

- **Data Imbalance** → Augmentation, oversampling & weighted loss
- **Heterogeneity** → Regridding & time synchronization
- **Data Gaps** → Temporal interpolation + IR/microwave fusion

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# IMPACT AND BENEFITS

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| Impact on Target Audience | Social & Economic Benefits | Environmental & Strategic |
|---|---|---|
| • **IMD & RSMC:** Faster cyclone detection, tracking & intensity analysis with reduced manual effort | • **Saves Lives:** Early warnings improve evacuation preparedness and reduce casualties | • **Climate Insight:** Long-term tracking of changing cyclone patterns |
| • **NDMA / SDMAs:** Earlier alerts support evacuation and relief planning | • **Community Safety:** Protects vulnerable coastal populations | • **Eco-Protection:** Early warnings can reduce exposure to spills and pollution risks |
| • **Coastal Communities & Fishermen:** Timely warnings provide more time to reach safety | • **Less Damage:** Advance action helps protect homes, crops, boats & infrastructure | • **Self-Reliance:** Leverages Indian satellite data and AI-based forecasting |
| • **Ports, Shipping & Aviation:** Early warnings support safer route and operational planning | • **Cost-Effective:** Uses satellite data and open-source AI tools | • **Global Goals:** Supports the Sendai Framework for Disaster Risk Reduction |
| • **Researchers:** Unified multi-satellite datasets support cyclone research | • **Better Planning:** Supports agriculture, insurance & disaster-relief allocation | • **Scalable:** Extendable across the Bay of Bengal, Arabian Sea & other hazards |

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# RESEARCH AND REFERENCES

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## ❖ Research Papers and Articles:

- **Satellite-based CNN intensity estimation** — Elsberry et al. (2019): deep learning for objective tropical-cyclone intensity estimation from satellite imagery. View paper
- **Deep learning + microwave observations** — Wimmers et al. (2019): DeepMicroNet demonstrates probabilistic TC intensity estimation from satellite microwave imagery. . View paper
- **Meteorological knowledge + AI** — Higa et al. (2021): domain knowledge improves satellite-image-based intensity classification. . Read research
- **Multispectral AI for cyclone intensity** — Liu et al. (2024): multispectral satellite information can improve deep-learning-based intensity estimation. Read research

## ❖ Authoritative Data & Standards:

- **ISRO MOSDAC — INSAT-3D/3DR** — Meteorological satellite products and documentation supporting multisource observation. MOSDAC references
- **IMD / RSMC New Delhi** — North Indian Ocean best-track data, cyclone monitoring, warning and forecast verification. IMD RSMC data
- **NOAA / NCEI — IBTrACS** — Global historical cyclone tracks and intensity records for training, validation and comparison. IBTrACS

